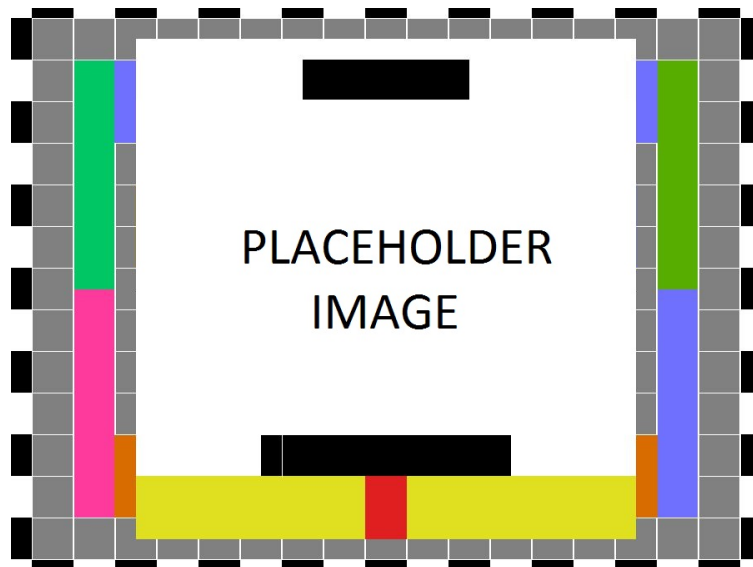


**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
THE UNIVERSITY OF TEXAS AT ARLINGTON**

**SYSTEM REQUIREMENTS SPECIFICATION
CSE 4316: SENIOR DESIGN I
FALL 2026**



**TEAM PERCEPTRON
DIGITDESK HANDWRITTEN FORM DIGITIZATION
SERVICE**

**ALAN TURING
GRACE HOPPER
JOHN VON NEUMANN**

ADA LOVELACE
CHARLES BABBAGE

REVISION HISTORY

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CONTENTS

1	Introduction	5
1.1	Purpose	5
1.2	Scope	5
1.3	Product Overview	5
1.4	Definitions, Acronyms, and Abbreviations	5
2	Requirements Engineering Approach	6
3	External Interface Requirements	7
3.1	EI-01: Ingestion API	7
4	Functional Requirements	7
4.1	FR-01: Field segmentation	7
4.2	FR-02: Digit recognition with confidence	8
4.3	FR-03: Confidence-based routing	8
4.4	FR-04: Human review	8
4.5	FR-05: Verified export & audit log	9
5	Usability Requirements	9
5.1	US-01: Efficient review	9
6	Performance Requirements	10
6.1	PE-01: Throughput & latency	10
6.2	PE-02: Field accuracy	10
7	Security & Privacy Requirements	11
7.1	SE-01: On-prem data handling	11
7.2	SE-02: Authenticated access	11
8	Maintenance & Support Requirements	12
8.1	MS-01: Model & data versioning	12
9	Other Quality Attributes	12
9.1	QA-01: Replaceable model	12
10	Verification & Traceability	13
11	Future Items	13
11.1	FI-01: Multi-template auto-detection	13

1 INTRODUCTION

This SRS specifies DigitDesk Handwritten Form Digitization Service, an on-premises service that digitizes handwritten numeric fields on scanned forms. It follows ISO/IEC/IEEE 29148:2018 [1].

1.1 PURPOSE

Agreed requirements baseline between Team Perceptron and the records office.

1.2 SCOPE

The system shall ingest scanned forms, segment numeric fields, recognize handwritten digits with calibrated confidence, auto-accept high-confidence results, route low-confidence fields to human review, and export verified structured data. Free-text (non-numeric) recognition is out of scope.

1.3 PRODUCT OVERVIEW

1.3.1 PRODUCT PERSPECTIVE

A containerized service on sponsor-provided on-prem infrastructure, exposing a REST API and a small review web UI; input is scanned images from the office's existing scanner workflow.

1.3.2 PRODUCT FUNCTIONS

Ingestion & preprocessing; field segmentation; digit recognition with confidence; confidence-based routing; human review; verified export; audit logging.

1.3.3 USER CHARACTERISTICS

Data-entry clerks (review UI users) and an IT administrator (deploys/monitors).

1.3.4 ASSUMPTIONS & DEPENDENCIES

Scans ≥ 200 dpi grayscale; consistent field layout per template; labeled validation set from the sponsor; on-prem GPU/CPU inference host.

1.4 DEFINITIONS, ACRONYMS, AND ABBREVIATIONS

Term	Definition
Field	A bounded region of a form holding one handwritten numeric value
Confidence	Calibrated probability the recognized value is correct
Review queue	Fields below the acceptance threshold, held for human check
IDP	Intelligent Document Processing

2 REQUIREMENTS ENGINEERING APPROACH

Identifiers [TYPE] - [NN] (EI, FR, US, PE, SE, MS, QA) with the 29148 attribute set (Description, Rationale, Source, Priority, Verification Method, Constraints, Applicable Standards). Requirements are singular and verifiable.

3 EXTERNAL INTERFACE REQUIREMENTS

3.1 EI-01: INGESTION API

DESCRIPTION

The system shall accept form images via an authenticated REST endpoint (POST /forms) as multipart PNG/TIFF and return a job identifier.

RATIONALE

The office's scan workflow must submit images programmatically.

SOURCE

Sponsor IT.

PRIORITY

Critical

VERIFICATION METHOD

Test — submit images, verify job id and status.

CONSTRAINTS

On-prem network only.

APPLICABLE STANDARDS

IETF RFC 9110 (HTTP), RFC 8259 (JSON).

4 FUNCTIONAL REQUIREMENTS

4.1 FR-01: FIELD SEGMENTATION

DESCRIPTION

For a known form template, the system shall locate each numeric field and produce a per-field digit-crop set.

RATIONALE

Recognition operates on isolated fields, not whole pages.

SOURCE

Team.

PRIORITY

Critical

VERIFICATION METHOD

Test — IoU of detected vs. ground-truth field boxes ≥ 0.9 .

CONSTRAINTS

Depends on consistent template layout.

APPLICABLE STANDARDS

None.

4.2 FR-02: DIGIT RECOGNITION WITH CONFIDENCE

DESCRIPTION

The system shall recognize the handwritten digit string in each field and attach a calibrated confidence to each field value.

RATIONALE

Confidence drives auto-accept vs. review routing.

SOURCE

Sponsor; team.

PRIORITY

Critical

VERIFICATION METHOD

Test — accuracy and calibration on the validation set.

CONSTRAINTS

Model runs on the on-prem host within latency budget (PE-01).

APPLICABLE STANDARDS

None.

4.3 FR-03: CONFIDENCE-BASED ROUTING

DESCRIPTION

Fields with confidence $\geq$ the configured threshold shall be auto-accepted; all others shall be placed in the review queue.

RATIONALE

Keeps a human check exactly where the model is unsure.

SOURCE

Sponsor.

PRIORITY

Critical

VERIFICATION METHOD

Demonstration — fields route per threshold.

CONSTRAINTS

Threshold configurable by admin.

APPLICABLE STANDARDS

None.

4.4 FR-04: HUMAN REVIEW

DESCRIPTION

A reviewer shall, through the web UI, see the field crop and proposed value, and confirm or correct it; the decision shall be recorded with the reviewer id.

RATIONALE

Auditable human oversight of uncertain fields.

SOURCE

Sponsor.

PRIORITY

Critical

VERIFICATION METHOD

Demonstration — review and correct a queued field.

CONSTRAINTS

Review UI runs in the office browser.

APPLICABLE STANDARDS

None.

4.5 FR-05: VERIFIED EXPORT & AUDIT LOG

DESCRIPTION

The system shall export verified records (form id, field, value, source: auto/reviewed) and retain an immutable audit log of every field decision.

RATIONALE

Downstream systems consume clean data; disputes require an audit trail.

SOURCE

Sponsor.

PRIORITY

High

VERIFICATION METHOD

Test — export matches decisions; log entries immutable.

CONSTRAINTS

Export format agreed (CSV/JSON).

APPLICABLE STANDARDS

None.

5 USABILITY REQUIREMENTS

5.1 US-01: EFFICIENT REVIEW

DESCRIPTION

A clerk shall review and clear a queued field in under 10 seconds median across 50 fields, using only keyboard confirm/correct.

RATIONALE

Review throughput determines the net labor saving.

SOURCE

Sponsor.

PRIORITY

High

VERIFICATION METHOD

Test — timed review sessions.

CONSTRAINTS

Readable crops in the UI.

APPLICABLE STANDARDS

WCAG 2.2 AA.

6 PERFORMANCE REQUIREMENTS

6.1 PE-01: THROUGHPUT & LATENCY

DESCRIPTION

The system shall process at least 5 pages/second and return per-page results within 2 s (95th percentile) on the on-prem host.

RATIONALE

Must keep up with the office scan volume.

SOURCE

Sponsor.

PRIORITY

High

VERIFICATION METHOD

Test — load test at target volume.

CONSTRAINTS

Bounded by host hardware.

APPLICABLE STANDARDS

None.

6.2 PE-02: FIELD ACCURACY

DESCRIPTION

Auto-accepted fields shall have $\leq 0.5\%$ error, and end-to-end field accuracy (after review) shall be $\geq 99.5\%$.

RATIONALE

Auto-accept must be safer than manual keying; the whole point is fewer errors.

SOURCE

Sponsor.

PRIORITY

Critical

VERIFICATION METHOD

Analysis — evaluate on the labeled validation set.

CONSTRAINTS

Requires representative validation data.

APPLICABLE STANDARDS

None.

7 SECURITY & PRIVACY REQUIREMENTS

7.1 SE-01: ON-PREM DATA HANDLING

DESCRIPTION

All form images and extracted data shall remain on sponsor-approved on-prem infrastructure; no data shall egress to external services.

RATIONALE

Form data may contain regulated PII.

SOURCE

Sponsor policy.

PRIORITY

Critical

VERIFICATION METHOD

Inspection — data-flow and egress review.

CONSTRAINTS

No cloud inference in prototype.

APPLICABLE STANDARDS

None.

7.2 SE-02: AUTHENTICATED ACCESS

DESCRIPTION

All API and review-UI access shall require authentication; every field decision shall be attributable to an identified user or the automated pipeline.

RATIONALE

Accountability and audit integrity.

SOURCE

Sponsor; team.

PRIORITY

High

VERIFICATION METHOD

Test — unauthenticated access refused; decisions attributed.

CONSTRAINTS

On-prem identity provider.

APPLICABLE STANDARDS

NIST SP 800-63B (guidance).

8 MAINTENANCE & SUPPORT REQUIREMENTS

8.1 MS-01: MODEL & DATA VERSIONING

DESCRIPTION

Each deployed model shall be identified by a version and the dataset hash it was trained on, recorded with every prediction it makes.

RATIONALE

Reproducibility and rollback when a model regresses.

SOURCE

Team.

PRIORITY

Moderate

VERIFICATION METHOD

Inspection.

CONSTRAINTS

Version metadata stored with results.

APPLICABLE STANDARDS

ISO/IEC 25010:2023 (analysability) [2].

9 OTHER QUALITY ATTRIBUTES

9.1 QA-01: REPLACEABLE MODEL

DESCRIPTION

The recognition model shall be replaceable via configuration without code changes to preprocessing, routing, or the API.

RATIONALE

Lets the model improve as more real data is labeled.

SOURCE

Team.

PRIORITY

Moderate

VERIFICATION METHOD

Inspection — interface review.

CONSTRAINTS

Defined model interface.

APPLICABLE STANDARDS

ISO/IEC 25010:2023 (modularity) [2].

10 VERIFICATION & TRACEABILITY

Req ID	Requirement (abbrev.)	Method	Acceptance
FR-02	Recognition + confidence	T	meets PE-02
FR-03	Confidence routing	D	routes per threshold
PE-01	Throughput/latency	T	≥ 5 pg/s, ≤ 2 s p95
PE-02	Field accuracy	A	auto err $\leq 0.5\%$; e2e $\geq 99.5\%$
SE-01	On-prem only	I	no egress observed

11 FUTURE ITEMS

11.1 FI-01: MULTI-TEMPLATE AUTO-DETECTION

DESCRIPTION

Automatic identification of the form template without pre-configuration.

RATIONALE

Broadens applicability but exceeds prototype scope.

SOURCE

Sponsor.

PRIORITY

Future

VERIFICATION METHOD

Test (if implemented).

CONSTRAINTS

Needs many labeled templates.

APPLICABLE STANDARDS

None.

REFERENCES

- [1] ISO/IEC/IEEE, “ISO/IEC/IEEE 29148:2018 — Systems and software engineering — Life cycle processes — Requirements engineering,” International Organization for Standardization, Geneva, Switzerland, Tech. Rep., 2018.

- [2] ISO/IEC, “ISO/IEC 25010:2023 — Systems and software engineering — Systems and software Quality Requirements and Evaluation (SQuaRE) — Product quality model,” International Organization for Standardization, Geneva, Switzerland, Tech. Rep., 2023.